# Calibration Is Necessary but Not Sufficient: Tail-Aware Prediction Market Forecasting


## Abstract

Prediction markets are often evaluated descriptively: a market is called well-calibrated when prices match realized frequencies on average. That view is useful but incomplete. Calibration is not a primitive property of a market; it is an outcome generated by liquidity, participation, timing, category structure, and the way information arrives. We propose a causal study of those drivers using a continuously collected Polymarket research stack. The local repository already supports market ingestion, trade aggregation, calibration tracking, market microstructure computation, and several causal-analysis modules. Our empirical strategy centers on resolved binary markets, deriving pre-resolution probabilities from trade histories and estimating how calibration varies with market structure, activity, and domain. The paper's central claim is that prediction-market calibration should be modeled as a causal outcome with heterogeneous mechanisms rather than as a single pooled descriptive statistic. In a first broad pass with a $\geq 10$ trade filter, the current pipeline yields 51,023 resolved binary markets, with final observed pre-resolution prices available for 45,506 of them. On that slice, final-price calibration is already strong in aggregate (Brier $= 0.0690$, $\mathrm{ECE}_{10} = 0.0196$), but the descriptive heterogeneity across categories is large enough to motivate a causal follow-up. We further argue that binary calibration is necessary but not sufficient in fat-tailed or asymmetric domains: when multiple threshold markets exist on the same event, the relevant object is the exceedance curve rather than a single threshold probability. A new ladder study on 1,144 distinct threshold ladders shows that this extension is already empirically available in the repo: 16.1% of ladders exhibit some adjacent-strike dominance violation and 5.2% exhibit a material (>2c) violation, while at-the-money contract accuracy has near-zero correlation with adjacent arbitrage edge ($r = 0.083$). The current draft is therefore grounded in implemented infrastructure and live empirics rather than in a purely speculative proposal.


## 1 Introduction

Prediction markets are increasingly treated as live probabilistic forecasts for politics, macroeconomics, sports, crypto, and science. Classical work shows that these markets can be reasonably calibrated, especially near resolution, but it also shows that calibration varies with horizon and market design [4]. More recent work suggests that calibration is structured and domain-specific rather than uniform across all markets [3]. What remains underdeveloped is a causal account of why some markets calibrate well while others fail.

That gap matters for both market science and AI forecasting. If a forecaster uses market prices as a baseline, it needs to know when the market should be trusted, when it should be discounted, and which observed market conditions are mechanisms rather than symptoms. A purely correlational treatment of spread, volume, or concentration is not enough. For actual trading systems, the target is even stricter: not just forecast quality, but deployable edge under Kelly sizing, portfolio limits, and tail-risk controls.

There is also a deeper objection. In thin-tailed settings, good calibration of a single probability forecast may already be close to the decision-relevant object. In fat-tailed or asymmetric settings, however, a well-calibrated threshold probability can still be a poor guide to expected loss, tail exposure, or right-tail opportunity. Taleb et al. argue that tournament-style probability assessment and tail-risk assessment can come apart sharply [7]. For nonnegative outcomes with exceedance curve $S(k) = P(X > k)$, the decision-relevant objects are integrals of that curve such as $E[X] = \int S(k)\,dk$ and $E[X^2] = 2\int kS(k)\,dk$, not merely one point on $S$. That critique is especially relevant when the forecast is later used inside a trading, hedging, or policy system.

This paper targets that gap. Our central question is:

> Which market-level and microstructural features causally drive prediction-market calibration, and how does that structure vary across domains?

The present draft adds a second question that follows directly from the first:

> When the underlying outcome is naturally thresholded at many levels, can we recover a useful exceedance curve rather than scoring only one binary contract?

**Contributions.**

1. We reframe prediction-market calibration as an outcome generated by market structure and microstructure rather than as a single descriptive statistic.
2. We define a repo-grounded extraction pipeline for resolved Polymarket markets using trade histories and metadata instead of the currently unreliable historical price table.
3. We specify a causal workflow that combines calibration slicing, structure learning, and effect estimation.
4. We operationalize ladder-level exceedance diagnostics, including truncated first- and second-moment proxies and a variance proxy recovered from Over/Under strike ladders.
5. We connect ladder incoherence to trading deployment: cross-strike dominance violations imply relative-value spreads and supply a natural risk-management label for Kelly haircuts or portfolio gating.

## 2 Repo-Grounded Data and Measurement

The drafting strategy is intentionally constrained by what is already implemented in the local `polymarket` repository. The codebase contains a continuous Polymarket ingestion pipeline, calibration primitives, microstructure computation jobs, and a family of causal-analysis modules that already implement PC-style structure learning, transfer-entropy analysis, interrupted time-series event-impact analysis, and manipulation detection.

The strongest current empirical substrate is the resolved binary market set. For each market, we define the *positive* event as the first listed outcome. Each trade is then mapped into a positive-event probability:

$$\hat{p}_t = \begin{cases} \texttt{price}_t & \text{if trade outcome} = \text{positive outcome} \\ 1 - \texttt{price}_t & \text{otherwise.} \end{cases}$$

This lets us reconstruct a probability path even when only one side of the binary market is actively traded.

Our contract-level outcome is Brier error at one or more horizons before resolution:

$$\text{Brier}_h = (\hat{p}_h - y)^2,$$

where $\hat{p}_h$ is the last observed implied probability before horizon $h$ and $y \in \{0, 1\}$ indicates whether the positive outcome ultimately wins [1]. We also compute log loss, Expected Calibration Error, and binned reliability curves.

**Why trade-derived probabilities?** The checked-in audit in the repo flags the historical `market_prices` table as unreliable for retrospective analysis because `condition_id` was empty in that snapshot. The first empirical pass therefore reconstructs pre-resolution prices from `market_trades`, which do contain `condition_id`, `outcome`, `price`, and `timestamp`.

**Current extraction footprint.** On April 14, 2026, the first calibration pass with a $\geq 10$ trade filter returned 51,023 resolved binary markets across 343 categories. Coverage is 45,506 markets at the final observed pre-resolution price, 35,488 at 6 hours before resolution, 25,231 at 24 hours, and 6,585 at 168 hours. Median trade count is 59 and median market duration is 4.30 days.

**Beyond single-threshold calibration.** A binary contract gives one exceedance probability, $P(X > k)$. For thin-tailed questions this may already be useful, but for fat-tailed or strongly asymmetric domains it is not enough: the decision-relevant object is closer to an exceedance curve, a tail integral, or a scenario-conditional distribution than to one threshold probability. In practical terms, this means Proposal A should not stop at "is this contract calibrated?" but should also ask whether the market provides enough threshold information to approximate more of the right tail.

When a ladder of thresholds $k_1 < \cdots < k_n$ is available, the exceedance curve can be converted into truncated moment proxies:

$$\widehat{E[X]} \approx k_1 + \sum_i S(k_i)(k_{i+1} - k_i), \qquad \widehat{E[X^2]} \approx k_1^2 + \sum_i S(k_i)(k_{i+1}^2 - k_i^2$$

which in turn give a truncated variance proxy $\widehat{\text{Var}}(X) = \widehat{E[X^2]} - \widehat{E[X]}^2$. A single threshold contract cannot recover these objects on its own. The same ladders can also be checked for dominance violations: if $P(X > k_{i+1}) > P(X > k_i)$ for $k_{i+1} > k_i$, then the ladder is not merely poorly calibrated but internally inconsistent in a way that matters for relative-value trading and portfolio risk control.

## 3 Feature Families

The paper separates broadly available market features from sparse high-frequency features.

**Primary feature families.** These are available for most resolved markets with sufficient trade data:

- market structure: total volume, duration, outcome count, negative-risk structure, category;
- trade activity: trade count, average trade size, maximum trade size, trade concentration, trades per day;
- temporal concentration: late-volume ratio and recent activity proxies;
- recent market movement: one-day and one-week price-change fields from market metadata.

**Secondary feature families.** These exist in the repo but are currently much sparser on resolved markets:

- bid-ask spread, effective spread, realized spread;
- order-book imbalance and depth metrics;
- wallet concentration and insider-score features;
- toxicity, price impact, and Kyle's lambda.

These features are scientifically important, but the current paper treats them as a subset analysis unless historical coverage is improved before submission.

## 4 Identification Strategy

The workshop memo proposed a four-stage pipeline. The current paper narrows that plan to what can be defended from the code and data already in hand.

**Stage 1: descriptive calibration decomposition.** We first quantify how calibration changes with category, trade intensity, volume regime, and horizon. This yields the first hard empirical result and motivates the causal question with visible heterogeneity.

**Stage 2: structure learning.** The repo already contains simplified PC-style graph discovery and complementary directional information-flow modules. The first submission therefore estimates a feature-level graph over observed market attributes and calibration outcomes, with category-specific re-estimation where sample sizes permit [5, 6].

**Stage 3: effect estimation.** After selecting interpretable treatments with reasonable overlap, we estimate their effect on calibration. The most realistic first-pass treatments are: high versus low trade intensity, high versus low concentration, short versus long time-to-resolution, and high versus low recent information-arrival regimes.

**Stage 4: validation.** Validation is practical rather than maximalist: temporal holdout by resolution date, category-specific robustness checks, and analyses with and without sparse high-frequency features.

**Distributional extension via ladders.** When Polymarket lists many Over/Under thresholds on the same event, the market is implicitly offering multiple samples from a survival curve. This creates a second empirical layer on top of contract-level calibration:

- monotonicity diagnostics for the implied exceedance curve;
- integrated Brier score across thresholds;
- truncated first- and second-moment proxies, plus a variance proxy;
- decoupling tests between at-the-money contract accuracy and full-ladder quality;
- adjacent-strike arbitrage labels for profitability and risk gating;
- eventual tail-shape tests on the right tail where enough strikes exist.

This is not yet full extreme-value theory, but it is a concrete way to move beyond probability-at-one-threshold and toward the object that matters for expected-value and tail-risk decisions.

## 5 Initial Empirical Status

The empirical work has started. The new script `experiments/run_cal` extracts resolved binary markets from ClickHouse, reconstructs positive-outcome probability paths from trades, and writes paper-facing artifacts: market-level CSVs, horizon-wise summary metrics, reliability bins, and slice tables by category, trade count, and volume quartile.

The first completed pass yields the following overall calibration summaries:

- final observed price: $n = 45{,}506$, Brier $= 0.0690$, log loss $= 0.2080$, $\mathrm{ECE}_{10} = 0.0196$;

- 6h horizon: $n = 35{,}488$, Brier $= 0.0896$, $\mathrm{ECE}_{10} = 0.0177$;
- 24h horizon: $n = 25{,}231$, Brier $= 0.0549$, $\mathrm{ECE}_{10} = 0.0193$;
- 168h horizon: $n = 6{,}585$, Brier $= 0.0799$, $\mathrm{ECE}_{10} = 0.0095$.

These horizon summaries are descriptive and are not directly comparable as time-to-resolution effects because each horizon is observed on a different coverage subset.

The key descriptive finding is heterogeneity. At the final-price horizon, Esports markets have Brier = 0.0230 on 8,763 contracts, while Sports markets have Brier = 0.1778 on 12,999 contracts; Politics sits between those extremes at Brier = 0.0577 on 844 contracts. Trade-count bins are much more stable, clustering between roughly 0.067 and 0.071 Brier, while the top volume quartile is noticeably worse than the bottom quartile (0.0855 vs. 0.0516).

These are still descriptive rather than causal results, but they already show that calibration is not homogeneous across the Polymarket universe. The code path now exists in the repo and runs directly on the live ClickHouse data backing the rest of the stack, which means the next step is no longer dataset construction but effect estimation and graph recovery.

**Threshold-ladder extension.** To address the limitation of single-contract calibration, we added a second script, `experiments/run_threshold_ladder_study.py`, that reuses the market-level calibration output and reconstructs Over/Under ladders from multiple thresholds on the same event. To avoid mixing unrelated props inside one event slug, ladders are grouped by normalized question template. On the current data snapshot, this returns 13,166 O/U markets and 1,144 distinct ladders with at least three thresholds. The median ladder has 4 thresholds and the mean has 4.33.

At the ladder level, the mean integrated final-price Brier score is 0.1843. The mean isotonic L1 repair gap is 0.0030, 16.1% of ladders exhibit at least one adjacent dominance violation, and 5.2% exhibit a material (>2c) violation. Most ladders remain too short to distinguish cleanly between exponential and power-law right-tail decay, but the infrastructure now exists to study exactly that question when richer strike ladders appear.

The stronger empirical point is deployability. The updated ladder script now extracts a central at-the-money contract for each ladder together with cross-strike arbitrage labels. Mean at-the-money Brier is 0.2004. It correlates strongly enough with full-ladder Brier to remain useful ($r = 0.679$), but it has almost no relationship with adjacent arbitrage gross edge ($r = 0.083$). Even among bottom-quartile at-the-money ladders, 31.5% still contain some adjacent dominance violation and 6.6% still contain a material (>2c) one. For trading systems, that means calibration alone is not the right deployment label: Kelly sizing and portfolio limits should also react to ladder coherence.

## 6 Limitations

The first submission must stay honest about data constraints.

1. Historical price reconstruction currently depends on trades rather than a clean dedicated price table.
2. Orderbook overlap for resolved markets is much thinner than trade overlap, so orderbook-heavy claims should remain subset-specific.
3. Wallet-derived participation features are missing for a substantial fraction of markets.
4. Category imbalance is severe; pooled estimates can be dominated by sports and esports unless explicitly stratified.
5. The threshold-ladder analysis is currently dominated by sports and esports totals, which are useful for methodology but not yet a direct test of catastrophic fat-tail forecasting.
6. The recovered moment and variance proxies are truncated to the observed strike range and are not yet normalized across categories with different outcome units.
7. Some large ladder violations likely reflect stale last-trade states in thin contracts, so observed relative-value spreads are a profitability label rather than a guarantee of executable arbitrage at scale.

## 7 Conclusion

Prediction-market calibration should not be treated as a single descriptive statistic detached from the mechanisms that generate it. The local Polymarket stack now makes it possible to study calibration as a market-level causal outcome, beginning with resolved binary markets and trade-derived implied probabilities. The immediate next step is to move from these descriptive calibration slices to effect estimation and graph recovery on the most plausible drivers of market quality. At the same time, the threshold-ladder results push Proposal A beyond plain calibration: when the underlying object of interest is tail-sensitive, the right target is not a single well-calibrated threshold probability but

a better estimate of the exceedance curve or its tail functionals. For Polymarket trading systems, the same lesson appears in more operational form: deployment should depend on ladder coherence and distributional quality, not only on one-contract calibration. A natural next step is to combine these diagnostics with scenario-conditioned forecasting questions of the kind used in recent AI economics worldbuilding exercises [2].

## References

[1] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[2] Convergence Analysis. Threshold 2030: Comprehensive summary. https://www.convergenceanalysis.org/threshold-2030/comprehensive-summary, 2024. Accessed April 14, 2026.

[3] Nam Anh Le. Decomposing crowd wisdom: Domain-specific calibration dynamics in prediction markets. *arXiv preprint arXiv:2602.19520*, 2026.

[4] Lionel Page and Robert T. Clemen. Do prediction markets produce well-calibrated probability forecasts? *The Economic Journal*, 123(568):491–513, 2013.

[5] Judea Pearl. *Causality*. Cambridge University Press, 2009.

[6] Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. MIT Press, 2000.

[7] Nassim Nicholas Taleb, Ron Richman, Marcos Carreira, and James Sharpe. The probability conflation: A reply. *arXiv preprint arXiv:2301.10985*, 2023.